Applied Energy 329 (2023) 120175

Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apenergy

Current-matching erases the anticipated performance gain of next-generation two-terminal Perovskite-Si tandem solar farms

M. Tahir Patel [a], Reza Asadpour [a], Jabir Bin Jahangir [a,*], M. Ryyan Khan [b], Muhammad A. Alam [a,*]

[a] Electrical and Computer Engineering Department, Purdue University, West Lafayette, IN, USA
[b] Department of Electrical and Electronic Engineering, East West University, Dhaka, Bangladesh

HIGHLIGHTS

- Global yield potentials of fixed tilt and tracking solar farms employing bifacial two-terminal tandem are quantified.
- Effective albedo collection determines the relative performance gain of bifacial 2T tandem systems.
- Only 3.5–5% gains at optimal albedo conditions are expected worldwide compared to constituent single junction HIT subcell.
- Deviation of effective albedo from 2T tandem's design albedo erases the gain in yield potential owing to current-matching.

ARTICLE INFO

Keywords:
Solar energy farms
Photovoltaics
HIT
Bifacial PV
Perovskite
Tandem
Two-terminal tandem
Tracking
Optimum design
Utility-scale PV

ABSTRACT

The bifacial gain of various optimally-tilted, and tracking bifacial farms based on single-junction PERC and HIT technologies are well established. The solar module technology is, however, evolving rapidly with the commercial development of two, three, and four-terminal mono and bifacial HIT-Perovskite tandem cells underway. Given the complexity of current-matching in two-terminal tandem cells and significant variation of the weather conditions across the world, one wonders if the benefits of fixed-tilt and tracking cells obtained for single-junction solar cells would remain for tandem solar cells. In this paper, we use a detailed illumination and temperature-dependent bifacial solar farm model (supported by a detailed physical model for bifacial HIT-Perovskite tandem cells) to show that (a) row-to-row shading in solar arrays significantly suppresses the effective albedo collection and thereby the two-terminal (2T) tandem cell efficiency and relative gain compared to an optimal bifacial HIT cell, (b) the global energy yield potential of fixed-tilted and solar-tracking topologies would improve by adopting a 2T tandem design at optimal albedo, with maximum gain arising for tracking farms, (c) the 2T tandem cell/modules (subcell bandgaps, thickness) must be optimized for maximum benefit, and (d) even a relatively small deviation from the optimum will negate all benefits. Our results will broaden the scope and understanding of the emerging tandem bifacial technology by demonstrating global trends in energy gain for worldwide deployment and the need for location-specific tailoring of the module design.

1. Introduction

A vastly increased energy supply is needed to support the projected population of ~10 billion people by 2050 [1]. Renewable energy sources, such as photovoltaics (PV), are expected to play an important role because it is relatively inexpensive, can be integrated with a variety of land types, and does minimal damage to the environment. Indeed, rapid innovation in solar cell, module, and farm technologies has supported the continued reduction in the levelized cost of energy (LCOE). The next-generation technologies are seeing rapid market expansion, e.g., bifacial modules are expected to have ~40 % market share by 2028 [2], whereas solar-tracking has been implemented by ~70 % of newly installed utility-scale PV systems since 2015 [3]. Further, the location-specific benefits of solar tracking with bifacial modules have already been quantified, and tracking, bifacial farms are likely to be deployed in many regions of the world [4].

To further reduce the LCOE, the PV industry is already looking ahead to developing the next-generation module technologies based on

* Corresponding authors.
E-mail addresses: jabir@purdue.edu (J. Bin Jahangir), alam@purdue.edu (M.A. Alam).

https://doi.org/10.1016/j.apenergy.2022.120175
Received 24 May 2022; Received in revised form 4 October 2022; Accepted 16 October 2022
Available online 20 November 2022
0306-2619/

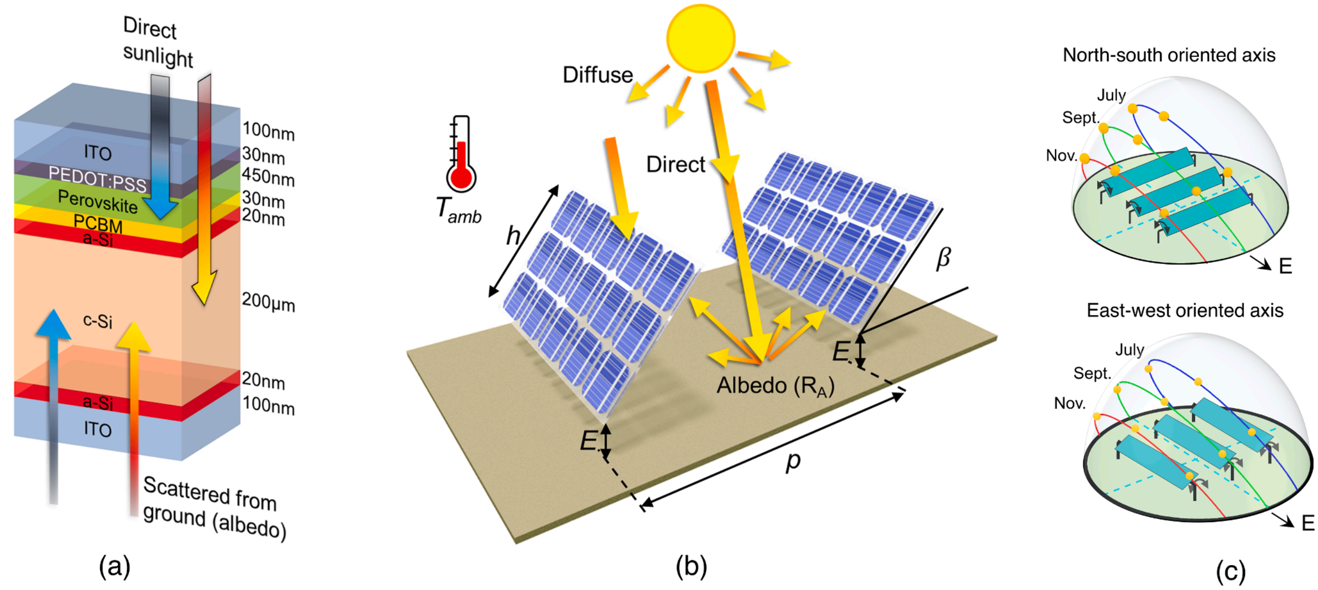

Fig. 1. PVK-HIT 2T tandem cell structure and the schematics of bifacial solar farms. (a) Two terminal (2T) tandem structure and thickness of each of the layers showing how the incident light is absorbed. (b) Various irradiance components and relevant design parameters of a bifacial solar array. (c) Schematics of single axis tracking bifacial solar farms.

monofacial and bifacial Perovskite/Silicon (PVK/Si), Perovskite/Perovskite (PVK/PVK), and Perovskite/Organic (PVK/BHJ) tandem solar cells. The efforts have been driven by two observations: the thermodynamic promise of remarkable efficiency gain for two-terminal (2T) tandem solar cells [5], and the fortuitous exact matching of the perovskite bandgap and silicon-heterojunction (HIT) solar cells to the ideal bandgap required by the thermodynamic limit [6] with a typical average albedo of 20–50 % across the world. Extraordinary photo-conversion efficiencies, e.g., 25.7 % [7], 27 % [8], 29 % [9], have been reported and initial reliability results appear promising. With the projected 20–30 % gain in energy yield [10], one wonders if the 2T tandem fixed-tilt and tracking farms would transform the renewable energy landscape.

The anticipated commercialization of 2T tandem solar cells has prompted several theoretical and experimental studies investigating the outdoor performance of PV systems utilizing these cells. Among them, several numerical studies thus far have predicted the energy yield potential of bifacial modules in a standalone or solitary array configuration for fixed-tilt and tracking configurations. A theoretical study by Dupre et al. [11] predicted 20 % energy gain with a fixed-tilt, standalone bifacial 2T tandem system compared to bifacial single-junction cells. Deploying the modules in a solar-tracking configuration leads to increased yield potential. Compared to fixed-tilt configuration, Schmager et al. predicted a 21–23 % gain in energy yield with single-axis-tracking (SAT) and a 32 % gain with a dual-axis-tracking system employing monofacial 2T tandem cells [12]. In another numerical study, Onno et al. predicted that a bifacial 2T tandem in a single-axis-tracking configuration produces 20 % more energy than a fixed-tilt configuration [13]. Recently, Babics et al. [14] reported the field performance of bifacial 2T tandem modules in desert conditions at Thuwal, Saudi Arabia. They found that a *standalone* module on an SAT configuration shows a 55 % gain in daily energy yield relative to a fixed-tilt configuration on a day around the summer solstice. However, solar modules in a farm are deployed in arrays and thus row-to-row shading plays an important role in determining the irradiation available to the rear surface of bifacial modules. Since with an increasing number of rows, the rear irradiation received by the center module in an array can drop by 15 % [15], a realistic system-level performance prediction must explicitly account for row-to-row shading. Therefore, optimistic gains suggested by standalone systems cannot be taken as a realistic guide for the performance gain of 2T tandem solar farms.

Furthermore, prior work on bifacial 2T tandem technologies has emphasized the critical importance of current matching in these technologies [16]. The bandgaps of the tandem cells can be optimized for an average albedo ($\langle R_A \rangle$), see Fig. 1(a). The 2T configuration outperforms the single-junction (SJ) counterparts over a relatively narrow albedo range ($\langle R_A \rangle \pm \Delta R_A$). At too low or too high albedo, the bottom and top cells respectively limit the current, suppressing the efficiency. For a solar farm shown in Fig. 1(b) and 1(c), the latitude- and time-dependent row-to-row shading makes the effective intercepted albedo ($R_A^*(t)$) a variable quantity. Unless $\langle R_A \rangle - \Delta R_A < R_A^*(t) \langle \langle R_A \rangle + \Delta R_A$, the 2T tandem farm will be less efficient and less reliable (due to Joule heating of the mismatched current) compared to SJ solar cells. Coupled with the temperature-dependent variation of the bandgaps, the advantages of 2T tandem fixed-tilt and tracking solar cells is thus by no means guaranteed. The purpose of this paper is to explore this trade-off carefully and quantify parameters that must be optimized for maximum energy gain in the context of a solar farm.

This work is the continuation of our series of studies on standalone and farm-level bifacial PV design considerations. Thus far, we have performed worldwide analyses investigating the optimum design parameters for energy maximization of a single bifacial PV module [17], a vertical bifacial PV farm [18], a ground-sculpted bifacial PV farm [19], and cost minimization of a fixed-tilt bifacial PV farm [20], and bifacial solar tracking farms [4]. These analyses have been performed with a solar farm model (see Fig. 2(a)) based on the co-optimization of optical-electrical-thermal considerations coupled with satellite-derived weather information. The solar farm model has been extensively validated by field measurements [21] and results published in the literature [17]. The model, when tested against field measurement, demonstrated good accuracy (2.7 % mean absolute error) in predicting the yield performance of a fixed-tilt bifacial system [21].

In this paper, we will use the thermodynamic analysis from ref. [6], physical electro-optical device simulation based on the model reported in ref. [22], and the solar-farm model integrated into a simulation framework for tandem systems (summarized in Fig. 2) to answer three fundamental questions:

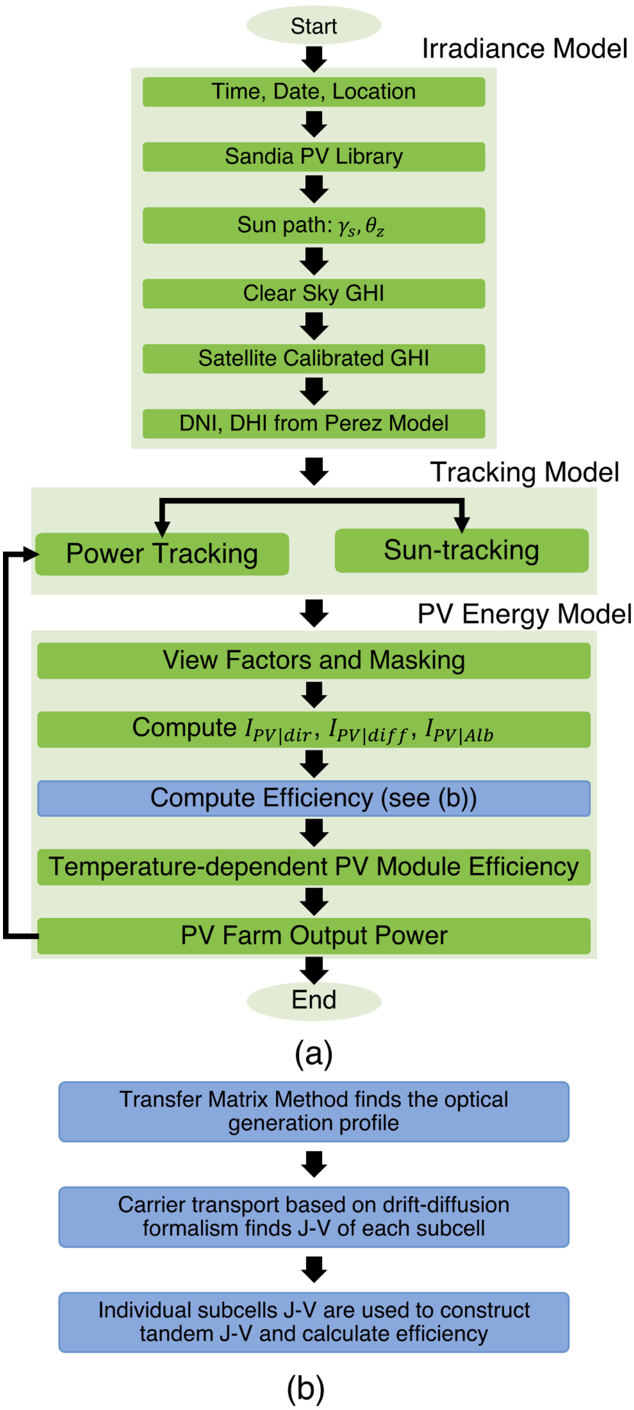

Fig. 2. The cell to farm simulation framework used in this paper. (a) The simulation framework integrates irradiance, light collection, and tracking models to calculate the energy yield of a bifacial tracking solar farm. (b) The device-modeling framework to calculate the current-matched 2T tandem cell efficiency. The temperature-corrections are made self-consistently in part (a) of the algorithm. For the equations, see Table A1.

a) Given latitude-dependent irradiance and temperature conditions, what locations in the world are particularly well-suited for the bifacial PVK-HIT tandem solar farms?
b) What is the average energy gain for fixed-tilt and solar tracking farms employing 2T tandem modules? How variable is the energy gain?
c) What conditions would ultimately dictate the choice of 2T tandem vs. HIT modules in a specific region?

The paper is organized as follows. The modeling framework is summarized in *Sec*. 2, the global energy yield potentials for fixed-tilt and tracking bifacial tandem modules are discussed in *Sec*. 3, and the insights regarding the viability of the 2T tandem PV are discussed in *Sec*. 4. Our conclusions and recommendations are summarized in *Sec*. 5.

2. Modeling framework

To model bifacial tandem fixed-tilt and tracking PV farms, one must generalize and integrate several sub-models developed in previous studies [23]. We define the solar module array in the farm through the following parameters: azimuth angle (γ_A) from the north, module height (h), module tilt angle (β), elevation from the ground (E), pitch (p), and ground albedo (R_A). The module efficiency in STC (η_{STC}) is solar intensity-dependent which is further corrected to operating temperature ($\eta(T_{cell})$). The various aspects of this model are discussed in Appendix A1. In the following, we briefly describe the key features of the model.

The simulation framework is summarized in Fig. 2(a). In our irradiance model, for any given location, we use the PVLIB toolbox [24] from Sandia National Laboratory to estimate the diurnal time-varying sun-path and clear-sky Global Horizontal Irradiance (GHI or I_{GHI}). The clear-sky GHI is scaled to a local weather-aware practical value using data from NASA POWER database [25]. The GHI is then split into Direct Normal Irradiance (DNI or I_{DNI}) and Diffuse Horizontal Irradiance (DHI or I_{DHI}) components using Orgill-Hollands-Perez model [17]. Next, at each time step, the spatial distribution of sunlight incident on the module's front and back surfaces, and on the ground are calculated. Treating direct light as parallel rays, shadows cast by the module onto adjacent modules, and on the ground are calculated according to the position of the sun. We use the view factor method to estimate the diffuse sunlight on the module faces and the ground [17]. The view factor method is also used to calculate the fraction of scattered light from the ground on the module. The light collection model is described in Appendix A1.1. The net light collection on module faces due to DNI and DHI collection is accounted for at each time step. In the case of single-axis solar tracking systems, the module tilt β is updated at each time-step about a fixed horizontal rotation axis for maximum module power output. The rotation axis can be oriented along the north–south or east–west direction, as shown in Fig. 1(c).

The light collected by the front and rear surfaces of the bifacial panel is estimated and then used to find the irradiance-dependent efficiency of the 2T tandem solar cell. The optimum bandgaps of subcells of a tandem solar cell depend on the albedo [6]. Here, we will optimize the tandem design for the typical average albedo of $R_A = 0.3$. Towards this goal, we need to calculate the cell-level characteristics (e.g., efficiency) as a function of incident sunlight. As shown in Fig. 3(a), at the cell level [22], a full-wave optical model, implemented by the transfer-matrix approach, gives absorption and photocarrier generation profiles within the thickness of the tandem or the HIT solar cell at a given incident sunlight intensity [26]. Subsequently, the position-resolved transport of photogenerated electron and hole concentrations inside the cell is simulated by a self-consistent solution of Poisson and continuity equations (see Table A3) by a commercial device simulator MEDICI [27] to find the corresponding subcell J-V characteristics shown in Fig. 3(b). Finally, the integrated cell efficiency (η_{STC}) for various combinations of front and rear irradiances are calculated by series combinations of the subcell J-V characteristics, as shown in Fig. 3(c) (see Appendix A1.3 for details).

Clearly, η_{STC} jointly varies with the module front and rear face's light intensity. To visualize the entire incident irradiance-efficiency landscape, we have let the rear irradiance vary between 0 and 1 sun, because in practice, the irradiances will depend on the array geometry and surface albedo. Then, the module output is calculated using a circuit model based on the cell efficiency and light collection while considering spatially non-uniform light and partial shading. The operating cell temperature T_{cell} and the relevant efficiency $\eta(T_{cell})$ is self-consistently

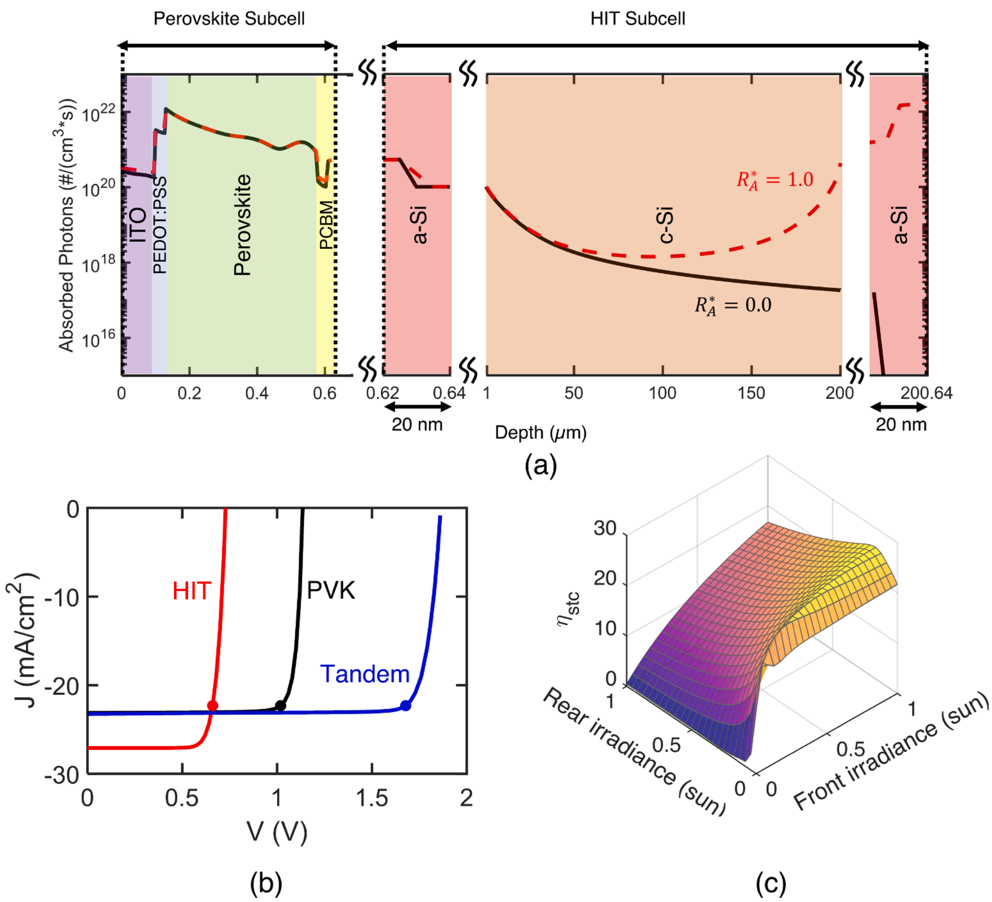

Fig. 3. Simulation of PVK-HIT 2T tandem cells. (a) Photo-generation profile of both subcells at (bifacial irradiance) $I_{front} = 1000W/m^2$ and $I_{rear} = 1000\text{W/m}^2$ (0W/m^2) shown by red dashed (black solid) line. (b) J-V characteristic of a tandem solar cell and subcells at full front irradiance ($I_{front} = 1000W/m^2$) and 30 % rear irradiance ($I_{rear} = 300\text{W/m}^2$). The circles show the V_{mp} and I_{mp} of the tandem cell and corresponding points in the subcells. (c) Efficiency of the 2T tandem solar cell at different front and rear irradiance. Efficiency is defined as P_{out}/P_{in} and $P_{in} = I_{front} + I_{rear}$ and 1 sun $= 1000\text{W/m}^2$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

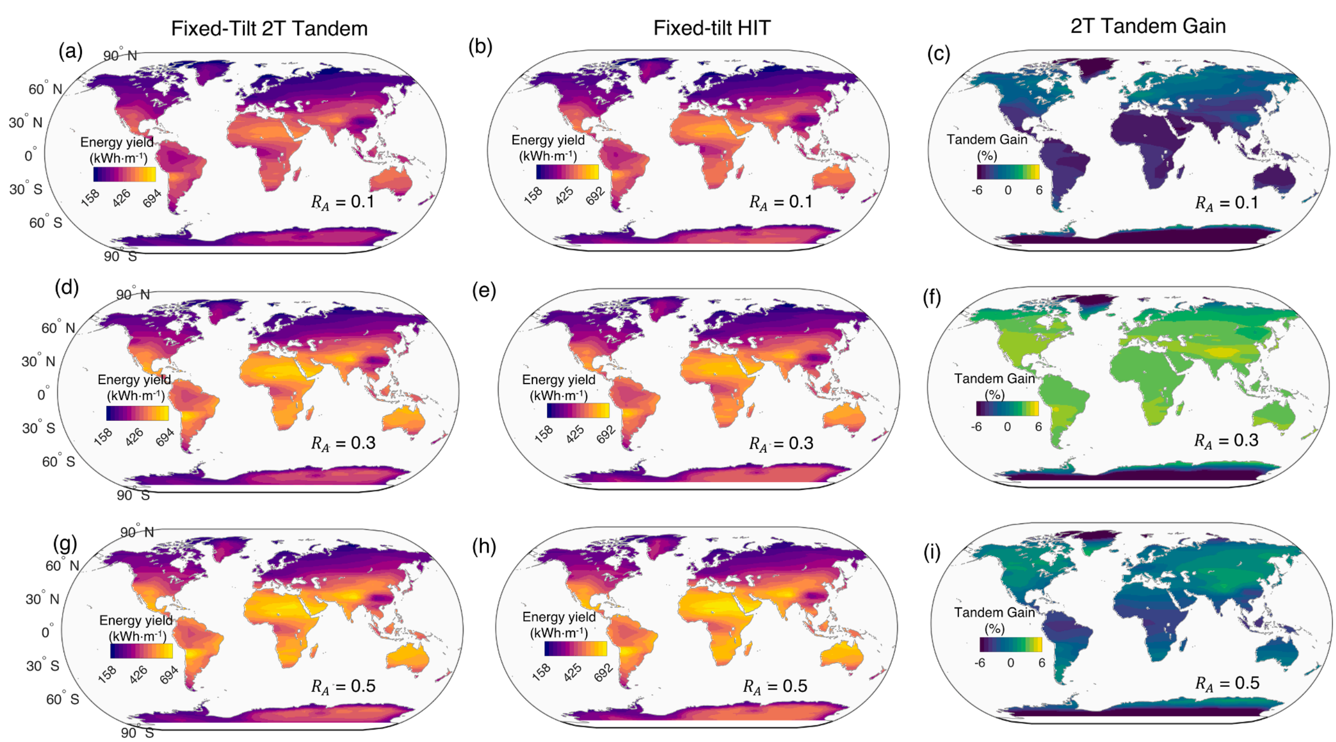

Fig. 4. Predicted global yearly energy yield potentials for 2T and HIT and relative energy gains of fixed-tilt configuration for ground albedo coefficients, $R_A = 0.1$ (a-c), 0.3 (d-f), and 0.5 (g-i). The tandem gain with 2T tandem is maximum only when the albedo condition meets its design albedo, $R_A = 0.3$ (f).

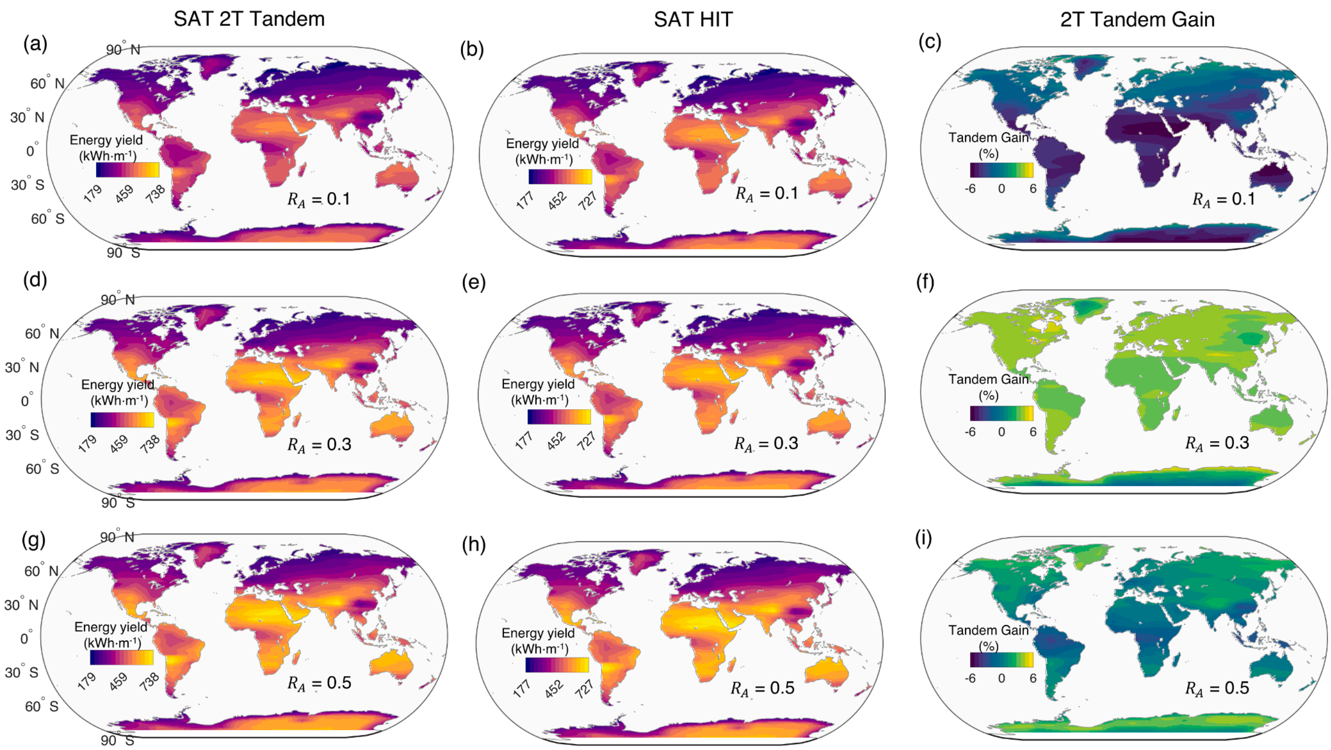

Fig. 5. Predicted global yearly energy yield potentials for 2T and HIT, and relative energy gains of single-axis tracking (SAT) configuration for ground albedo coefficients, $R_A = 0.1$ (a-c), 0.3 (d-f), and 0.5 (g-i). Similar to the fixed-tilt configurations, the tandem gain with 2T tandem is maximum only when the albedo condition meets its design albedo, $R_A = 0.3$ (f).

calculated while evaluating the module output. For a module with tandem cells, each subcell has its temperature coefficient and therefore requires a non-trivial approach to calculate the effective T_{cell} and temperature-dependent efficiency — the details are discussed in Appendix A1.4. Finally, the power output is integrated to estimate the yearly energy yield (YY) of the solar farm.

3. Global yield potential and expected gain of bifacial two-terminal tandem solar farms

Utilizing the detailed opto-thermal-electrical model described above, we carried out a global simulation of bifacial solar farms comprising of

(i) fixed-tilt PVK/HIT 2T tandem modules,
(ii) single-axis tracking (SAT) PVK-HIT 2T tandem modules, and
(iii) bifacial single junction HIT modules as reference.

Since the energy yield potential of bifacial systems improves with increasing ground albedo coefficient, the yield potentials are computed for three albedo coefficients typical of different natural surfaces, namely $R_A = 0.1$, 0.3 and 0.5.

3.1. Energy yield of fixed tilt farms: 2T tandem vs HIT

In Fig. 4, we compare the global yearly energy yield potential of north–south-facing, fixed tilt bifacial 2T tandem (column 1) and HIT (column 2) solar farms for different albedo conditions. Column 3 plots the location-specific relative gain of 2T tandem over HIT cell. The calculations make two assumptions: (a) for computational efficiency, the optimum module tilt angle is calculated by an empirical formula from ref. [17] and (b) a fixed $p/h = 3$ is assumed optimal for every location. The energy yield is computed per row length of array in kWh $\cdot$ m^{-1}

By focusing from the top to the bottom rows of Fig. 4, we find that the energy yield increases monotonically with increasing R_A for both bifacial technologies. As shown in Fig. 4(a), when $R_A = 0.1$, 2T tandem's yearly energy yield is predicted to range from 158 to 557 kWh $\cdot$ m^{-1}. Higher yields are expected at locations near the equator (within $\pm 40^{\circ}$) experiencing higher insolation. In comparison, bifacial HIT's yield potential ranges from 159 kWh $\cdot$ m^{-1} at high latitudes to 581 kWh $\cdot$ m^{-1} near the equator. Thus, the energy yield of 2T tandem at low albedo is comparable to (or lower than) that of HIT cells: Fig. 4(c) shows that the relative gain is negative for most regions of the world, with only a negligible gain at higher latitudes (e.g., northern Europe) owing to increased diffuse light collection enhancing bottom cell current. It is not surprising that the 2T tandem cell optimized for $R_A = 0.3$ underperforms at $R_A = 0.1$. This is because, at such low albedo conditions, the current in the bottom cells of the 2T tandem becomes the limiting factor.

With a 20 % increased ground albedo coefficient ($R_A = 0.3$), the energy yield potential increases significantly for each technology. When $R_A = 0.3$, the 2T tandem energy yield, as shown in Fig. 4(d), ranges from 172 to 664 kWh $\cdot$ m^{-1}. In comparison, the HIT's energy yield ranges between 173 and 636 kWh $\cdot$ m^{-1} as shown in Fig. 4(e). Evidently as shown in Fig. 4(f), for this albedo condition the 2T tandem outperforms bifacial HIT everywhere in the world within $\pm 60^{\circ}$ latitude with gains ranging between ~2 and 5 %.

A further 20 % increased albedo ($R_A = 0.5$) further increases the energy yield of systems employing both technologies. It can be observed from Fig. 4(g) and (h) that both 2T tandem and HIT show similar energy yield potential: the potential of 2T tandem ranges between 179 and 694 kWh $\cdot$ m^{-1} and HIT's 188 to 692 kWh $\cdot$ m^{-1}. As shown in Fig. 4(i), while globally relative gains of 2T tandem range between approximately -4% to 2 %, at most locations near the equator 2T tandem now underperforms compared to HIT. Therefore, the relative gains in energy yields of farms employing bifacial 2T tandem cells (engineered for $R_A = 0.3$) are strongly dependent on the ground surface condition of the farms. The bifacial 2T tandem only offers a positive gain globally when the PV system's ground albedo coefficient matches the albedo the 2T tandem cells were designed for. We will further explain this key insight in *Sec.* 4.

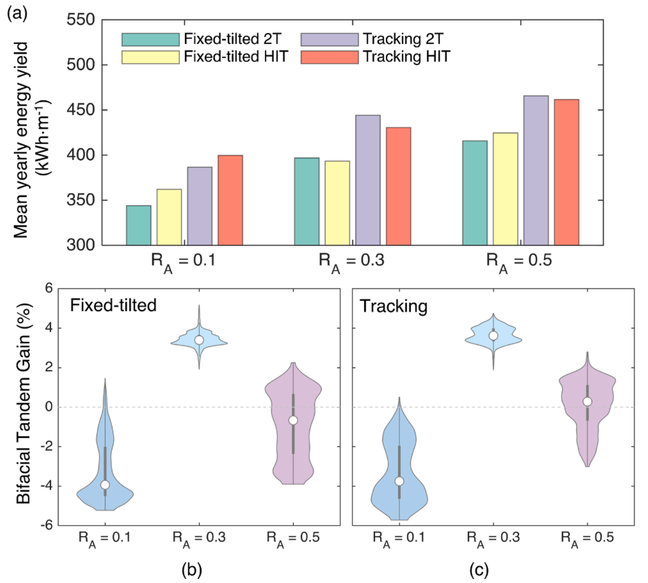

Fig. 6. Mean yield potential and relative gains. (a) A summary of estimated global mean yearly yield potentials for bifacial 2T tandem and HIT deployed in fixed-tilt and tracking configuration as a function of ground albedo coefficient (R_A). 2T tandem outperforms HIT when R_A is close to 0.3, the design albedo. Distribution of estimated global gains in energy yield with bifacial 2T tandem technology relative to bifacial HIT as function of R_A for (b) fixed tilt and (c) tracking configurations. Here, we only consider gains observed between $\pm 60^{\circ}$ latitude. The gain is maximized for both configurations when R_A is 0.3 and shows a narrow distribution. The gain falls off as R_A moves away from 0.3 and broader variability is observed.

3.2. Energy yield of tracking farms: 2T tandem vs HIT

Next, we predict the performance of solar farms employing a single-axis tracking configuration. Similar to the previous figure for fixed-tilt solar farms, Fig. 5 shows the global yearly yield potentials of solar-tracking bifacial 2T tandem and HIT farms for the three ground albedo conditions. Here we present the global yield potential maps of SAT systems with an east–west oriented rotation axis. Additionally, the yield potentials and relative gains of systems with a north–south oriented axis are presented in the supplementary Fig. S5. The time-dependent optimum β was determined through a power-tracking algorithm described in Appendix A1.2. Again, we set $p/h = 3$ for every location.

Focusing on the first row of Fig. 5, we observe that , when $R_A = 0.1$, the yearly energy yield of bifacial tandem with tracking configuration ranges from 179 kWh $\cdot$ m^{-1} at higher latitudes (e.g., northern Europe) to 595 kWh $\cdot$ m^{-1} at locations near the equator. In comparison, the energy yield for a bifacial HIT farm ranges between 178 kWh $\cdot$ m^{-1} to 623 kWh $\cdot$ m^{-1}. Similar to the results for fixed-tilt configurations, for this sub-optimal albedo condition, tracking HIT outperforms tracking 2T tandem farms in most locations. Although the relative energy yield gains for 2T tandem can be slightly positive ($\sim$1%) at higher latitudes, it remains negative in most parts of the world.

As shown in Fig. 5(d), with a 20 % increased albedo coefficient ($R_A = 0.3$) the yield of the tracking 2T tandem configuration increases and ranges between 199 and 701 kWh $\cdot$ m^{-1}. In comparison, tracking HIT's yield potential ranges between 194 and 674 kWh $\cdot$ m^{-1} as shown in Fig. 5(e). Similar to the fixed-tilt case, we find that the 2T tandem's energy gain is positive ($\sim$2–5 %) for all locations with $R_A = 0.\ 3$ as shown in Fig. 5(f). With a further 20 % increased albedo ($R_A = 0.5$), 2T tandem's yield potential improves and ranges between 209 and 737 kWh $\cdot$ m^{-1} as shown in Fig. 5(g). However, farms employing bifacial HIT also demonstrate similar yield potentials ranging between 210 and 726 kWh $\cdot$ m^{-1} as shown in Fig. 5(h). As shown in Fig. 5(i), the relative gains in yield potential of 2T tandem farms range from approximately -3 to 2.8 %: 2T tandem produces up to 3 % lower yield at locations near the equator and only shows a small positive gain at higher latitudes, e.g., in northern Canada. Consequently, similar to fixed-tilt configuration, for both lower ($R_A = 0.1$) or higher ($R_A = 0.5$) albedo conditions, a single-junction HIT cell is preferred over the 2T tandem (bandgaps optimized at $R_A = 0.3$) at most locations between $\pm 40^{\circ}$ latitude to produce greater energy yield. These results again underscore that the 2T tandem cells must be optimized for relevant local albedo conditions for optimal yield performance.

3.3. 2T tandem vs HIT yield potentials summarized

Fig. 6 summarizes the absolute and relative gain results from Figs. 4 and 5. Fig. 6 (a) shows that the mean yearly energy yield increases for both technologies with an increase in the albedo coefficient, as expected. It also shows that tracking configurations produce the greatest yields irrespective of the albedo condition. We find that tracking about an east–west oriented axis on average produces 10.15 % and 13.46 % more yield compared to the fixed-tilt configuration for HIT and 2T tandem farms, respectively. On the other hand, tracking about a north–south

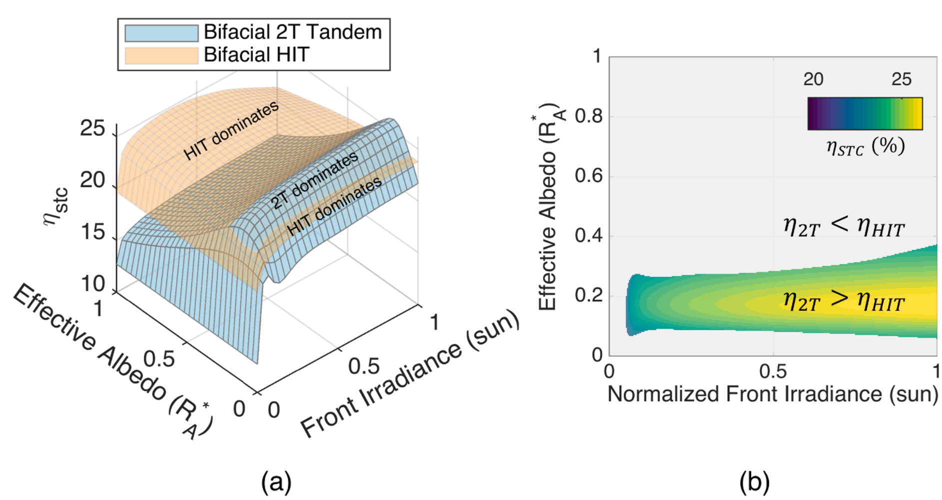

Fig. 7. (a) STC efficiency of bifacial HIT and 2T tandem solar cells as function of normalized front irradiance and effective albedo. (b) Positive relative gain in energy obtained for 2T tandem when it has higher efficiency compared to HIT. 2T dominates only for a range of effective albedo for which optimal current matching between subcells is achieved. Relative gain is maximized when $R_A^* = R_{A,max-gain}^* \sim 0.17$.

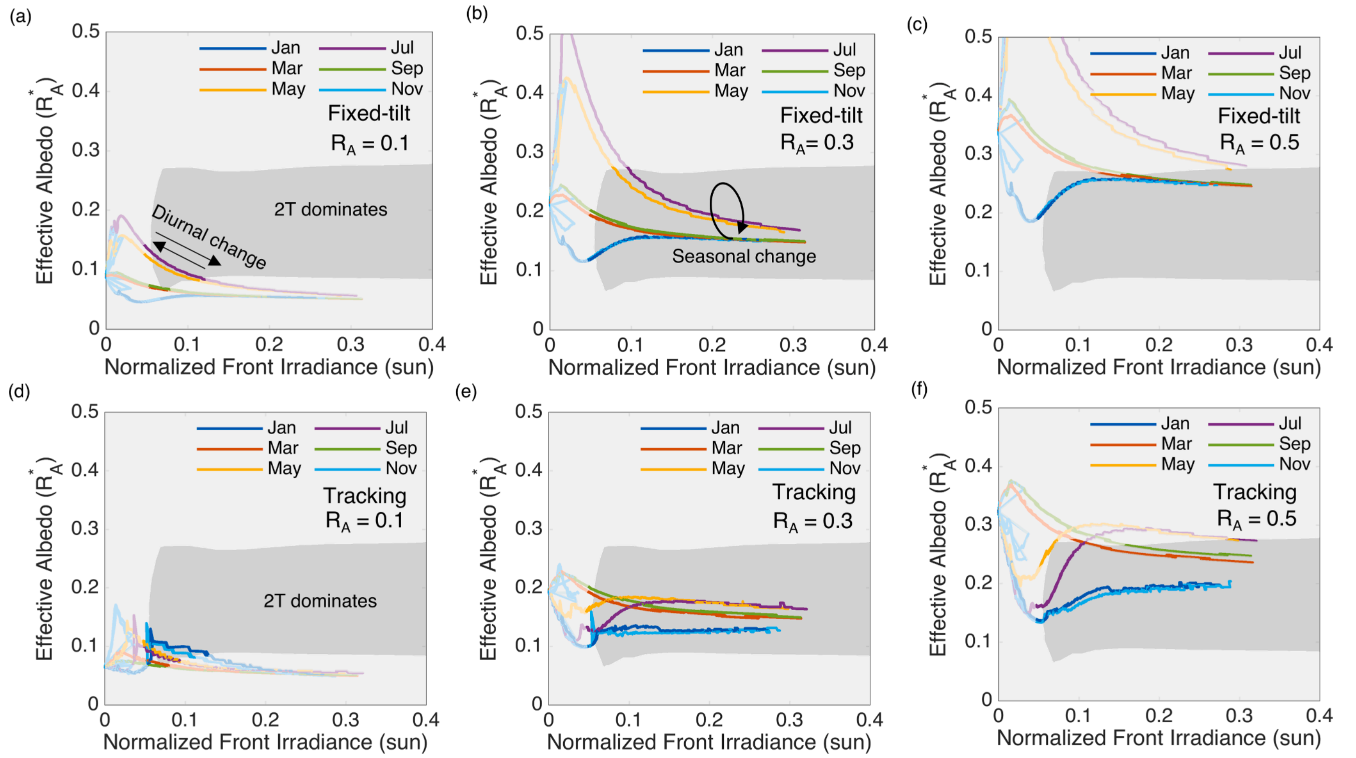

Fig. 8. Daily evolution of effective albedo and front irradiance for (a-c) fixed-tilt configuration and (d-f) east–west-axis tracking configuration during different months in an example site in Algeria for $R_A = 0.1$, 0.3 and 0.5. The gray-shaded area signifies the combination of front irradiance and effective albedo that results in larger efficiency for bifacial 2T-tandem due to optimal current-matching condition. With $R_A = 0.3$ (b and e), optimal current matching between subcells is achieved and 2T tandem outperforms HIT; however, the relative gains diminish as ground albedo deviates from the optimal.

oriented axis leads to a higher absolute energy yield as expected with a mean gain of 15.43 % and 15.59 % compared to a fixed-tilt solar farm. These gains are lower than those reported for standalone systems. Fig. 6 (b) and (c) show the distributions of tandem's gain for fixed-tilt and tracking configurations, respectively. Here, we only consider gains obtained at locations within $\pm 60^{\circ}$ latitudes where most solar farms are situated. These plots, in addition to highlighting the similarity of energy gain trends for both fixed-tilt and tracking farms as a function of R_A, show that the 2T tandem's gains for both configurations are maximized at$R_A = 0.3$: *an average gain of 4 % with a relatively narrow distribution worldwide is expected.* For non-optimal albedo, i.e., R_A= 0.1 or 0.5, a broader variability in 2T tandem performance is observed.

This broader variability can be understood by considering the spatial distribution of 2T tandem gain shown in Figs. 4 and 5. We observed that, for non-optimal albedo conditions, while at most locations near the equator (within latitudes $\pm 40^{\circ}$) 2T tandem gain is negative, the gain improves with increasing latitudes. (The variation in tandem gain as a function of latitude is shown in Supplementary Fig. S4.) The locations at higher latitudes have a higher contribution from DHI to total irradiance allowing for increased diffused light collection by the rear surface. This in turn leads to enhanced bottom cell current in 2T tandem despite a non-optimal ground albedo. Hence, the relative gain of the 2T tandem is not solely determined by the albedo condition, but also by the array geometry. We will further elaborate on this point in the following section.

In summary, variation of the ground albedo away from the value for which the tandem was engineered erases all the potential gain for the 2T tandem technology. This follows from the current-matching constraint in the series-connected 2T tandem cells.

4. Why is the relative gain of bifacial two-terminal tandem farms limited?

Although ground albedo coefficient (R_A) partly determines the fraction of irradiance available to the rear surface, the actual collection also depends on the module's geometric arrangement, i.e., irradiances at the front and the rear planes of the array (POA). Solar modules in a farm are deployed in periodic arrays that cast shadows on the ground underneath. This considerably reduces the light collected at the rear surface of a module placed within an array. Moreover, for series-connected 2T tandem cells, front and rear surface irradiances determine the currents in the top and bottom subcells, and their proportion dictates output power and efficiency. Thus, we define effective albedo (R_A^*) as the ratio of the rear and front POA irradiances incident on the module faces. Thus, while the ground albedo coefficient is a property of the surface underneath the array, the effective albedo (R_A^*) is a property of the PV system and depends on array geometry. The effective albedo can be seen as the ground albedo coefficient corrected for losses due to periodic shading in a solar farm.

Fig. 7 depicts the evolution of STC efficiencies (η_{STC}) of bifacial 2T tandem and bifacial HIT with respect to incident front-face irradiance (I_{Front}) and effective albedo R_A^*. The light collected by the rear face (I_{Rear}) is then equal to $I_{Front} \times R_A^*$. Fig. 7(a) shows that the efficiency of bifacial 2T tandem surpasses that of bifacial HIT only under certain illumination conditions. When front-face irradiance is rather low (<0.05 sun) in the early morning or late afternoon, the efficiency of the 2T tandem is greatly diminished by subcell current mismatch owing to a smaller current in the HIT bottom cell. Then with increasing front and rear irradiances in the middle of the day, the efficiency of the 2T tandem increases. Specifically, the bifacial 2T tandem has higher efficiency than bifacial HIT when $0.07 < R_A^* < 0.27$ and sufficient front irradiance (>0.05 sun) is available; the maximum 2T tandem efficiency is obtained for $R^*_{A,max-gain} \approx 0.17$. This region is highlighted in Fig. 7(b). Only under such illumination conditions, the 2T tandem shows a positive relative gain. Beyond $R^*_A \sim 0.27$, the two-terminal tandem's output power is bounded by the PVK top cell's smaller current ceiling. Consequently, a further increase in effective albedo or increased input power translates into decreased efficiency of 2T tandem. In contrast, the output efficiency of the bifacial HIT cell (shown overlaid on Fig. 7(a)) is not subject to the current-matching constraint and does not degrade drastically like 2T tandem.

Next we illustrate how the choice of ground albedo coefficient for a given array geometry influences the relative gain of solar farms employing 2T tandem cells. Fig. 8 shows effective albedo vs front irradiance characteristics of fixed-tilt and east–west axis tracking bifacial tandem farms for the three ground albedo conditions ($R_A = 0.1$, 0.3, 0.5). A location in Algeria ($23.45^\circ N, 10^\circ E$) was chosen arbitrarily for this example. In each figure, an individual line plot shows the evolution of R^*_A and I_{Front} throughout a single day. Six days from six months were chosen to delineate the change in this farm characteristic owing to changes in sun path throughout the year. The daily variations of R^*_A as a function of time during these months are shown in Supplementary Fig. S1. We observe that R^*_A has a higher value in the mornings (when front irradiance is low) and gradually settles down as the day progresses, i.e., front irradiance increases. Like Fig. 7(b), the shaded regions represent the illumination condition where the 2T has higher efficiency compared to bifacial HIT.

Fixed-tilt Solar Farms. Fig. 8(a) shows that although $R_A = 0.1$, the effective albedo (R^*_A) is lower than 0.1 during most of the day throughout the year. This small light collection by the bottom cell limits the current in bifacial two-terminal tandem and thus, results in smaller relative gains for 2T tandem. This also explains why in most places near the equator ($<\pm 40^\circ$ latitude) bifacial tandem underperforms (see Fig. 4 (c) and Fig. 5(c)).

Increasing the ground albedo coefficient to 0.3 (see Fig. 8(b)) increases the effective albedo to 0.15–0.25 for the most part of the days. This increase in albedo collection increases the bottom cell current for the series 2T configuration, and the 2T efficiency dramatically improves and surpasses that of bifacial HIT by up to 5 % (also see Fig. 7). A further increase of the ground albedo coefficient to 0.5 (Fig. 8(c)) further increases the effective albedo collection; however, this increase in the light collection at the bottom cell cannot improve power output which will be limited by PVK top cell's lower current ceiling. We find that for $R_A = 0.5$ the efficiencies of the 2T tandem and HIT are close which results in the marginal gains ($\sim \pm 2\%$) observed across the globe. It follows that subsequent increases in ground albedo increase bifacial HIT's efficiency but reduces the 2T efficiency. The evolution of 2T tandem and HIT efficiencies vs time is shown in Supplementary Fig. S2.

Tracking Solar Farms. Similar effective albedo vs front irradiance characteristics is observed for east–west-axis tracking configuration as well, as shown in Fig. 8(d-f). As shown in Fig. 8(d), solar tracking alone is unable to compensate for the lack of rear POA irradiance due to small R_A. Accordingly, the 2T tandem's performance is limited by the bottom cell and shows a relative loss in gain compared to HIT. For $R_A = 0.3$, the effective albedo lies in the zone where the 2T tandem demonstrates positive gain compared to HIT throughout the year, as shown in Fig. 8 (e). For $R_A = 0.5$ shown in Fig. 8(f), however, we find that effective albedo during winter and autumn months (e.g., January, November) is lower than that in the fixed tilted system and has values close to $R_{A,max-gain}$. As such, 2T tandem produces a greater yield than HIT during the winter/autumn months but less energy during the summer months (see Supplementary Fig. S3). Since the sun is being tracked about an east–west oriented axis, during summer the module tilt angle is smaller compared to winter/autumn and consequently smaller fraction of the ground underneath is shaded by rows. Thus, the rear POA irradiance is increased during the summer months; however, 2T tandem performance is limited by the top cell. Overall, the gain in yearly yield of 2T tandem at this location is negligible (0.13 %). The results for a north–south-axis tracking system are shown in Supplementary Fig. S7.

A corollary from these results in Fig. 8 is that, in an array configuration, the effective albedo R^*_A can be significantly lower than the ground albedo coefficient R_A due to row shadings on the ground. For very large array pitch, or in the case of standalone systems, $R_A \approx R^*_A$. Therefore, conclusions and expected optimistic gains from standalone systems (e.g., ref. [14]) cannot be readily extrapolated to module arrays.

Finally, the above results show that the local albedo collection plays an important role in determining the viability of 2T tandem farms. The results underscore the need for location-specific farm design that accounts for the local albedo condition and the various seasonal factors that may affect it (e.g., an increase in albedo coefficient due to snowfall). To ensure maximal yield for a given 2T tandem module, the ground surface albedo should be engineered to match the optimal albedo, and the module should be placed in a tracking configuration.

5. Summary and conclusions

In this paper, we have discussed the design considerations of 2T tandem solar cells installed in fixed-tilt and tracking solar farms. We also explained how high-performance bifacial tandem solar farms design strongly depends on current-matching conditions. Our analysis leads to three key conclusions.

First, our results demonstrate that the design of the 2T tandem cells must be optimized for specific ground albedo. Assuming tandem optimized at a global mean albedo of 0.3, it is possible to obtain 3.5–5 % energy gain across all locations in the world. However, if the module design is not commensurate with the ground-albedo, the current-matching constraints will erase all gain – indeed, single-junction solar modules (e.g., HIT cells) are universally preferred over 2T tandem cells in such cases. Further, we found that, although the tracking systems with a north–south axis produce higher absolute yields for both technologies as expected, the relative gains of 2T tandem are slightly smaller for these systems. Thus, the conclusions of only modest gain of tandem solar farms compared to single-junction bifacial solar farms are independent of the specific farm topology considered.

Second, it would be difficult to manufacture 2T tandem cells optimized for a variety of albedo. Instead of relying on natural surface albedo, therefore, it may be more practical to engineer the ground surface so that the albedo is matched to the designed solar module. The additional expenses associated with engineering the ground surface albedo and the 2T tandem cells must therefore be carefully balanced against the energy gain obtained by using these next-generation solar modules in fixed-tilt or tracking configuration. Indeed, the farm-level energy gain reported in this paper will help refine the techno-economic analysis of tandem solar cells that have thus far relied on idealized energy gain under constant illumination and constant temperature conditions [28].

Third, some reports in the literature highlight the performance improvement of the 2T tandem systems by comparing the energy yield of fixed tilt vs tracking configuration. Evidently, tracking would improve energy yield for all technologies. To properly gauge the benefit of a technology, it should be compared to alternative technologies with the same topology (e.g., tracking HIT vs tracking 2T tandem). The results reported here follow this approach.

Finally, many of the challenges associated with 2T tandem configurations are obviated with three-terminal (3T) or four-terminal (4T) tandem configurations, which may eventually become the preferred alternative to the two-terminal-based designs being developed by various companies. The energy gain associated with the 3T and 4T tandem configurations will be a topic of future research.

CRediT authorship contribution statement

M. Tahir Patel: Methodology, Software. **Reza Asadpour:** Methodology, Writing – original draft. **Jabir Bin Jahangir:** Visualization, Software, Formal analysis, Writing – original draft. **M. Ryyan Khan:** Visualization, Writing – review & editing. **M.A. Alam:** Conceptualization, Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

We gratefully acknowledge the discussion with Profs. Peter Bermel, Rakesh Agrawal, and Mark Lundstrom from Purdue University.

Appendix A1. : Farm model details

The solar farm model consists of optical, thermal, and electrical submodels developed in previous studies [23]. To simulate the tandem structure, the previous models have been appropriately generalized. The following sections describe the various submodels depicted in Fig. 2: (1) irradiance and light collection model, (2) solar-tracker model, (3) optical simulation of tandem structure, and (4) temperature-dependence model of PVK-HIT tandem.

A1.1 Irradiance and light collection models

As shown in Fig. 1(b), we model a solar farm with the following design parameters: azimuth angle (γ_A) from the North; module height (h); module angle (β); elevation from the ground (E); pitch (p); and ground albedo (R_A).

We use the ***irradiance model*** and estimate the ground illumination for a particular time of the day with a time step of 1 min at a location specified by its latitude and longitude. We follow the modeling described in [23]. We use the PV library from Sandia National Laboratory to calculate the sun's trajectory (zenith (θ_Z) and azimuth angle (γ_S)) and the irradiance [18]. The Global Horizontal Irradiance (GHI or I_{GHI}) is ideally given by Haurwitz clear sky model [29]. Renormalizing this irradiance based on the NASA Surface meteorology and Solar Energy database [25] yields the local variation in GHI. The 22-year monthly average GHI data from the NASA POWER dataset has a spatial resolution of $1 \times 1^\circ$ (latitude $\times$ longitude). The GHI is then split into direct normal irradiance (DNI or I_{DNI}) and diffuse horizontal irradiance (DHI or I_{DHI}) components using Orgill and Hollands model [30]. Perez model [31] further improves the accuracy by estimating both the circumsolar and isotropic components of diffuse light.

We quantify the amount of ***light collected by the solar modules*** installed at that location. The modules have height h, are tilted at an angle β, separated by pitch (or period)p, and are oriented at an array azimuth angle $\gamma_A = 180^\circ$ (i.e., south-facing modules) for farms in the northern hemisphere and $\gamma_A = 0^\circ$ (i.e., north-facing modules) for farms in the southern hemisphere or $\gamma_A = 90^\circ$ for modules facing E-W direction. The collection of light on modules from the three components of irradiance (i.e., direct, diffuse, and albedo) are formulated separately and analyzed accordingly. Our approach to modeling the collection of light involves the view-factor calculation as described in ref. [17]. The corresponding equations are summarized in Table A1. A correctly implemented view-factor model gives identical results compared to the ray-tracing approach [32]. The solar farm simulation model is implemented in MATLAB..

Table A1
Equations associated with the optical model

(A.1)$I_{GHI} = I_{DNI} \times \cos(\theta_Z) + I_{DHI}$	(A.6)$I^{F,Panel}_{PV:Alb.diff}(l) = I_{Gnd:DHI} \times R_A \times F_{dl-gnd}(E,l) \times \eta_{diff}$
(A.2)$I^{F,Farm}_{PV:DNI}(l) = \begin{cases} I_{DNI}cos\theta_F(1 - R(\theta_F))\eta_{dir} & if \quad l\ is\ unshaded \\ 0 & if \quad l\ is\ shaded \end{cases}$	(A.7)$I^{F,Farm}_{PV:Alb.diff} = 1/h \int_0^h I^{F,Panel}_{PV:Alb.diff}(E,l)dl$
(A.3)$I^{F,Farm}_{PV:DHI} = I_{DHI}\eta_{diff}/h \int_0^h \frac{1}{2}(1 + \cos(\psi(E,l) + \beta))dl$	(A.8)$I^{Farm}_{PV:Alb} = I^{Farm}_{PV:Alb.dir} + I^{Farm}_{PV:Alb.diff}$
(A.4)$I^{F,Panel}_{PV:Alb.dir}(l) = I_{Gnd:DNI} \times R_A \times F_{dl-gnd}(E,l) \times \eta_{diff}$	(A.9)$I^{Total}_{PV(T)} = I^{Farm}_{PV:DNI} + I^{Farm}_{PV:DHI} + I^{Farm}_{PV:Alb}$
(A.5)$I^{F,Farm}_{PV:Alb.dir} = 1/h \int_0^h I^{F,Panel}_{PV:Alb.dir}(E,l)dl$	(A.10)$YY_T(p,\beta,h,E,\gamma_A,R_A) = \int_0^1 I^{Total}_{PV(T)}(p,\beta,h,E,\gamma_A,R_A)dY$

Table A2
Equations associated with the thermal model.

(A.11)$\eta(T_{cell}) = \eta_{STC}(1 - TC(T_{cell} - T_{STC}))$	(A.14)$V_{(mp,T)} = V_{mp,STC}(1 - TC(T_{cell} - T_{STC}))$
(A.12)$T_{cell} = T_{amb} + c_T\frac{\tau\alpha}{u_L}P_{POA}\left(1 - \frac{\eta(T_{Cell})}{\tau\alpha}\right)$	(A.15)$TC^* = (I_{mp1}V_{mp1}TC_1 + I_{mp2}V_{mp2}TC_2)/(I_{mp2}V_{mp2} + I_{mp1}V_{mp1}$
(A.13)$P_{POA} = I^{Farm}_{PV:DNI}/\eta_{dir} + (I^{Farm}_{PV:DHI} + I^{Farm}_{PV:Alb})/\eta_{diff}$	(A.16)$V_{mp,tandem,2T} = V_{mp,tandem}(1 - TC_{tandem}(T_{cell} - T_{STC}))$

A1.2 Tracking model

In a PV tracking system, instead of a fixed module tilt, the tilt angle $\beta(t)$ varies with time. The **power-tracking** model optimizes the time-dependent tilt angles to achieve maximum power at each time step—this will also maximize daily energy. The optimization is governed by the combined effects of (a) relative contributions of instantaneous DHI and DNI, (b) position of the sun, and (c) module-to-module shading. This power-tracking is targeted toward maximizing output, not the light collection. Thus, during the early and late part of the day, the system will compromise light collection to avoid shading loss and maximize power output.

A1.3 Optical simulation for tandem structure

The estimated light collected, from the previous section, on the front and rear surfaces of the solar module is used to find the intensity-dependent efficiency of the tandem solar cell. We begin by calculating the tandem cell efficiency for front and rear illumination values, from 0 to 1000 W/m^2, at standard testing conditions (STC). First, we calculate the position-resolved optical absorption in different layers of both perovskite (PVK) and HIT subcells by the full-wave solution of Maxwell's equations with the input of AM 1.5G illumination [33]. The materials in different layers of the cell are characterized by absorption coefficient and refractive indices that are obtained from the literature. Transfer matrix method (TMM) [26] calculations are used for the optical studies of the planar cell structure. In this approach, the entire solar cell stack, including the contact layers, is modeled using a series of interface and phase matrices. The central quantity which is calculated in this approach is the point-wise optical absorbance (A(λ, r)) inside various layers of the cell. The wavelength range of 300–1500 nm has been used for our calculations. The spatially resolved absorption profile is integrated over the wavelength range to create the generation profile for the electron and holes for self-consistent carrier transport simulation, as described below.

The transport of charged carriers (electrons and holes) is modeled by generalized drift–diffusion formalism [27] (see Table A3). Photo-generation is calculated from the optical absorption profile (integrated over the wavelengths) discussed previously. The electron and hole transport inside the cell is simulated by a self-consistent solution of Poisson and continuity equations by a commercial-grade device simulator MEDICI [34]. The generation term in the e–h continuity equations is calculated from the solution of the photo-generated profile. The recombination term in continuity consists of direct and Shockley-Read-Hall (SRH) recombination with lifetime found in the literature. We do not account for hot electron effects. The material properties used for electrical simulation can be found in Table A4 for the HIT subcell and Table A5 for the perovskite subcell.

We then find the efficiency based on the J-V curve resulting from the solution of drift–diffusion formalism. For example, as shown in Fig. 3, we

Table A3
Equations for carrier transport calculations.

(A.17) Poisson Equation: $\epsilon_r\epsilon_0\nabla^2\psi = -q(n_h - n_e)$	(A.19) Continuity Equation: $\nabla J_{e,h} = \left(G_{e,h} - R_{e,h}(n_e, n_h)\right)$
(A.18) Drift-Diffusion: $J_{e,h} = \mu_{e,h} n_{e,h}(-\nabla\psi) \pm D_{e,h}\nabla n_{e,h}$	(A.20) Recombination: $R_{e,h}(n_e, n_h) = B\left(n_e n_h - n_i^2\right) + \frac{n_e n_h - n_i^2}{\tau(n_e + n_h)}$

Table A4
The simulation parameters used in the numerical simulation of HIT CELL [22].

Properties	p-layer	i-layer	c-Si	i-layer	*n*-layer
Thickness	10 nm	10 nm	200μm	10 nm	10 nm
Doping (cm^{-3})	$N_A = 5\times10^{19}$	–	$N_D = 5\times10^{15}$	–	$N_D = 5\times10^{19}$
Hole mobility (cm^2/Vs)	0.2	2	330	2	2
Electron mobility (cm^2/Vs)	1	20	1030	20	10
Hole lifetime (s)	–	–	5×10^{-4}	–	–
Electron lifetime (s)	–	–	5×10^{-4}	–	–
Band gap (eV)	1.7	1.7	1.12	1.7	1.7
Electron affinities (eV)	3.9	3.9	4.05	3.9	3.9
Contact properties	Ohmic contacts, $s_f = 10^7 cm/s$ for both contacts				

Table A5
The simulation parameters used in the numerical simulation of Perovskite cell.

Properties	PEDOT: PSS	Perovskite	PCBM
Thickness	30 nm	450 nm	30 nm
Doping (cm^{-3})	$N_A = 5\times10^{17}$	$N_A = 1\times10^{16}$	$N_D = 3\times10^{17}$
Hole mobility (cm^2/Vs)	9×10^{-3}	4 0	1×10^{-2}
Electron mobility (cm^2/Vs)	9×10^{-3}	40	1×10^{-2}
Hole lifetime (s)	1×10^{-6}	80×10^{-9}	1×10^{-6}

(*continued on next page*)

perform an optical simulation with a front irradiance of $1000\ \mathrm{W/m^2}$ (i.e., 100 % of standard $1000\ \mathrm{W/m^2}$) and rear irradiance of $1000\ \mathrm{W/m^2}$). The top PVK subcell, having a thickness of 610 nm (including the ITO layer), absorbs the front illumination partially and generates 24 $\mathrm{mA/cm^2}$. The bottom HIT subcell, having a thickness of ~200 μm absorbs the transmitted front irradiance partially and the rear irradiance generates 14 (front irradiance) + 42 (rear irradiance) = 56 $\mathrm{mA/cm^2}$ current density. Based on the respective thicknesses and absorption coefficients of the top and bottom cells the current through both the cells is limited by the cell which generates lower current comparatively since they are in a series electrical connection. By adding up the voltages of the two subcells at the same current density, we construct the two-terminal tandem J-V characteristics. Here, the two-terminal tandem efficiency was defined as the output power divided by the total illumination intensity.

A1.4 Tandem thermal model with an effective temperature coefficient

The subsequent step in the modeling framework is to incorporate the ***temperature-dependent efficiency model.*** This model requires the collected light intensity, tandem efficiency at STC, and ambient temperature as inputs and yields effective efficiency as output [35]. The temperature-dependent efficiency loss involves a complex interplay of increased absorption due to bandgap reduction vs increased dark-current and reduced mobility. At practical illumination intensity, it is well-known that the efficiency $\eta(T)$ scales linearly with the module/cell temperature, see Eq. (A.11) in Table A2 where the rate/slope of degradation is given by the absolute temperature coefficient (TC) [36]. Note that, in a single junction/single material solar cell, the absolute TC has only one value. However, in a multi-junction/tandem solar cell, we need to consider temperature coefficients for the tandem subcells and calculate an effective value of temperature coefficient (TC*). Eq. (A.12), similar to Eq. (A.11), describes a linear relationship between voltage at maximum power point (V_{mp}) and temperature with an associated TC. In an electrically series connected 2T tandem cell, the total voltage ($V_{mp,tandem}$) across the solar cell at maximum power point is the sum of the voltages across individual subcells at maximum power point, i.e.,

$$V_{mp,tandem} = V_{mp,1} + V_{mp,2}$$

In the thermodynamic limit, V_{mp} remains constant with a varying incident light (irradiance) [5] Using this relationship between voltages, we can find out the effective temperature coefficient for the tandem cell given by Eq. (A.14). In Eq. (A.14) we have $TC_1 = TC_{PVK} = -0.379\frac{\%}{K}, TC_2 = TC_{HIT} = -0.258\frac{\%}{K}$ [37]. The total voltage across the tandem cell is given by Eq. (A.16). Here, we make two assumptions: (i) The subcells of the tandem solar cell are in thermal equilibrium, i.e., they have the same temperature ($T_1 = T_2 = T_{tandem}$), and (ii) the temperature-coefficient for the maximum power point current (I_{mp}) is negligible as compared to that of V_{mp}. Since $\eta = V_{mp} \times I_{mp}$ therefore,

$$TC(\eta) = TC(V_{mp}) + TC(I_{mp})$$

Thus, we substitute TC^* as the effective temperature-coefficient for the tandem cell in Eq. (A.11). Next, we solve Eq. (A.11) and (A.13) analytically using the ambient temperature (T_{amb}) and total light collected (P_{POA}) to find the temperature-dependent efficiency ($\eta_{T,tandem}$) of the tandem solar cell.

A1.5 Power and energy of a tandem bifacial PV farm

Finally, we find ***the daily, monthly, and yearly energy output*** of the farm. Using equations (A.2), (A.3), and (A.8) in Table A1, we arrive at Eq. (A.9) to find the time-varying spatially distributed light collection on the modules. This information combined with the thermal model (summarized in Table A2) is used in the circuit model to find the equivalent power generation. The circuit model includes the bypass diode model to take into account the partial shading of the PV modules and their effect on the PV performance [38]. To estimate energy output, we integrate the power generated over the desired period of time. We define the energy yield per pitch of a farm over one year as the yearly yield (YY) given by Eq. (A.10).

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apenergy.2022.120175.

References

[1] A. Parant. Les perspectives demographiques mondiales. 1990. doi: 10.3406/tiers.1983.4290.
[2] ITRPV. International Technology Roadmap for Photovoltaic (ITRPV): Results 2018 Including Maturity Report, Itrpv. 2019:76. Available from: https://itrpv.vdma.org/en/ueber-uns.
[3] Fu R, Feldman D, Margolis R. 110. U.S. Solar Photovoltaic System Cost Benchmark: Q1 2018. Nrel 2018:1–47. https://www.nrel.gov/docs/fy19osti/72399.pdfl.
[4] Patel MT, Ahmed MS, Imran H, Butt NZ, Khan MR, Alam MA. Global analysis of next-generation utility-scale PV: tracking bifacial solar farms. Appl Energy 2021;290:116478. https://doi.org/10.1016/j.apenergy.2021.116478.
[5] Alam MA, Ryyan Khan M. Shockley-Queisser triangle predicts the thermodynamic efficiency limits of arbitrarily complex multijunction bifacial solar cells. Proc Natl Acad Sci U S A 2019;116:23966–71. https://doi.org/10.1073/pnas.1910745116.
[6] Ryyan Khan M, Alam MA. Thermodynamic limit of bifacial double-junction tandem solar cells. Appl Phys Lett 2015;107(22):223502.
[7] Hou Y, Aydin E, De Bastiani M, Xiao C, Isikgor FH, Xue DJ, et al. Efficient tandem solar cells with solution-processed perovskite on textured crystalline silicon. Science (80-.) 2020;367:1135–40. doi: 10.1126/SCIENCE.AAY0262.
[8] Xu J, Boyd CC, Yu ZJ, Palmstrom AF, Witter DJ, Larson BW, et al. Triple-halide wide-band gap perovskites with suppressed phase segregation for efficient tandems. Science (80-.) 2020;367:1097–104. doi: 10.1126/SCIENCE.AAZ4639.
[9] Al-Ashouri A, Köhnen E, Li B, Magomedov A, Hempel H, Caprioglio P, et al. Albrecht, Monolithic perovskite/silicon tandem solar cell with >29% efficiency by enhanced hole extraction. Science (80-.) 2020;370:1300–9. doi: 10.1126/SCIENCE.ABD4016/SUPPL_FILE/ABD4016_AL-ASHOURI_SM.PDF.
[10] Lehr J, Langenhorst M, Schmager R, Gota F, Kirner S, Lemmer U, et al. Energy yield of bifacial textured perovskite/silicon tandem photovoltaic modules. Sol Energy Mater Sol Cells 2020;208:110367. https://doi.org/10.1016/j.solmat.2019.110367.
[11] Dupre O, Tuomiranta A, Jeangros Q, Boccard M, Alet PJ, Ballif C. Design rules to fully benefit from bifaciality in two-terminal perovskite/silicon tandem solar cells. IEEE J Photovoltaics 2020;10:714–21. https://doi.org/10.1109/JPHOTOV.2020.2973453.
[12] Schmager R, Langenhorst M, Lehr J, Lemmer U, Richards BS, Paetzold UW. Methodology of energy yield modelling of perovskite-based multi-junction photovoltaics. Opt Express 2019;27:A507. https://doi.org/10.1364/oe.27.00a507.
[13] Onno A, Rodkey N, Asgharzadeh A, Manzoor S, Yu ZJ, Toor F, et al. Predicted power output of silicon-based bifacial tandem photovoltaic systems. Joule 2020;4:580–96. https://doi.org/10.1016/j.joule.2019.12.017.
[14] Babics M, De Bastiani M, Balawi AH, Ugur E, Aydin E, Subbiah AS, et al. Unleashing the full power of perovskite/silicon tandem modules with solar trackers. ACS Energy Lett 2022;7(5):1604–10.
[15] R. IEA-PVPS, Bifacial Photovoltaic Modules and Systems: Experience and Results from International Research and Pilot Applications 2021 Task 13 Performance, Operation and Reliability of Photovoltaic Systems PVPS, 2021. www.iea-pvps.org (accessed April 15, 2022).
[16] Chung H, Sun X, Mohite AD, Singh R, Kumar L, Alam MA, et al. Modeling and designing multilayer 2D perovskite / silicon bifacial tandem photovoltaics for high efficiencies and long-term stability. Opt Express 2017;25:A311. https://doi.org/10.1364/OE.25.00A311.
[17] Sun X, Khan MR, Deline C, Alam MA. Optimization and performance of bifacial solar modules: a global perspective. Appl Energy 2018;212:1601–10. https://doi.org/10.1016/j.apenergy.2017.12.041.

[18] Khan MR, Hanna A, Sun X, Alam MA. Vertical bifacial solar farms: physics, design, and global optimization. Appl Energy 2017;206:240–8. https://doi.org/10.1016/j.apenergy.2017.08.042.
[19] Khan MR, Sakr E, Sun X, Bermel P, Alam MA. Ground sculpting to enhance energy yield of vertical bifacial solar farms. Appl Energy 2019;241:592–8. https://doi.org/10.1016/j.apenergy.2019.01.168.
[20] Patel MT, Khan MR, Sun X, Alam MA. A worldwide cost-based design and optimization of tilted bifacial solar farms. Appl Energy 2019;247:467–79. https://doi.org/10.1016/j.apenergy.2019.03.150.
[21] Bin Jahangir J, Al-Mahmud M, Shakir MSS, Haque A, Alam MA, Khan MR. A critical analysis of bifacial solar farm configurations: theory and experiments. IEEE Access 2022;10:47726–40. https://doi.org/10.1109/ACCESS.2022.3170044.
[22] Asadpour R, Chavali RVK, Ryyan Khan M, Alam MA. Bifacial Si heterojunction-perovskite organic-inorganic tandem to produce highly efficient (ηT* ∼33%) solar cell. Appl Phys Lett 2015;106. https://doi.org/10.1063/1.4922375.
[23] Khan MR, Patel MT, Asadpour R, Imran H, Butt NZ, Alam MA. A review of next generation bifacial solar farms: Predictive modeling of energy yield, economics, and reliability. J Phys D Appl Phys 2021;54(32):323001. https://doi.org/10.1088/1361-6463/abfce5.
[24] Holmgren WF, Hansen CW, Mikofski MA. Pvlib Python: a Python package for modeling solar energy systems. J Open Source Softw 2018;3:884. https://doi.org/10.21105/joss.00884.
[25] POWER. Surface meteorology and solar energy: a renewable energy resource web site (release 6.0); 2017. Available from: https://eosweb.larc.nasa.gov/cgi-bin/sse/sse.cgi? (n.d.).
[26] Pettersson LAA, Roman LS, Inganäs O. Modeling photocurrent action spectra of photovoltaic devices based on organic thin films. J Appl Phys 1999;86:487–96. https://doi.org/10.1063/1.370757.
[27] Pierret RF. Advanced semiconductor fundamentals. 2 ed. Upper Saddle River, N.J.: Prentice Hall; 1987.
[28] Zafoschnig LA, Nold S, Goldschmidt JC. The race for lowest costs of electricity production: techno-economic analysis of silicon, perovskite and tandem solar cells. IEEE J Photovoltaics 2020;10:1632–41. https://doi.org/10.1109/JPHOTOV.2020.3024739.
[29] Haurwitz B. Insolation in relation to cloud type. J Meteorol 1946;3:123–4. https://doi.org/10.1175/1520-0469(1946)003<0123:iirtct>2.0.co;2.
[30] Duffie JA, Beckman WA, McGowan J. Solar engineering of thermal processes. Am J Phys 1985;53(4).
[31] Perez R, Seals R, Ineichen P, Stewart R, Menicucci D. A new simplified version of the perez diffuse irradiance model for tilted surfaces. Sol Energy 1987;39:221–31. https://doi.org/10.1016/S0038-092X(87)80031-2.
[32] Berrian D, Libal J. A comparison of ray tracing and view factor simulations of locally resolved rear irradiance with the experimental values. Prog Photovoltaics Res Appl 2020;28:609–20. https://doi.org/10.1002/pip.3261.
[33] Tsai H, Asadpour R, Blancon JC, Stoumpos CC, Even J, Ajayan PM, et al. Design principles for electronic charge transport in solution-processed vertically stacked 2D perovskite quantum wells. Nat Commun 2018;9:1–9. https://doi.org/10.1038/s41467-018-04430-2.
[34] V. W-, Taurus Medici, (2004).
[35] Patel MT, Vijayan RA, Asadpour R, Varadharajaperumal M, Khan MR, Alam MA. Temperature-dependent energy gain of bifacial PV farms: a global perspective. Appl Energy 2020;276:115405. https://doi.org/10.1016/j.apenergy.2020.115405.
[36] Green MA. Solar cells: operating principles, technology, and system applications / Martin A. Green, (1982) xiv, 274 p. Available from: https://books.google.com/books/about/Solar_Cells.html?id=ODAEAQAAIAAJ (accessed May 17, 2022).
[37] Dupré O, Vaillon R, Green MA. Thermal behavior of photovoltaic devices. 2017. doi: 10.1007/978-3-319-49457-9.
[38] Deline C, Dobos A, Janzou S, Meydbray J, Donovan M. A simplified model of uniform shading in large photovoltaic arrays. Sol Energy 2013;96:274–82. https://doi.org/10.1016/j.solener.2013.07.008.

Glossary of the symbols and notations

Parameters: **Definition**
TC: Temperature Coefficient (%/° C)
TC:* Effective temperature coefficient
T_{amb}: Ambient Temperature (° C)
T_{cell}: Module Temperature (° C)
η_{STC}: Power conversion efficiency at Standard Testing Condition (%)
c_T: Correction term used for the daily average temperature data
τ: Transmittance of glazing (no unit)
α: Absorbed fraction (no unit)
θ_Z: Zenith angle (*degrees*)
γ_S: Sun azimuth (*degrees*)
θ_F: Angle of incidence at the front face of the panel
p: Row-to-row distance between the bottom edges of consecutive arrays (m)
h: Height of the panel (m)
I: Irradiance (kW/m^2)
E: Elevation (m)
β: Tilt angle (degrees)
γ_A: Azimuth angle of the array measured from the north (degrees)
R_A: Albedo
YY: Yearly energy yield per array row length (kWh/m)
R_A^*: Effective albedo
Sub/Super-scripts: **Definition**
a: Ambient
Alb: Albedo
DHI: Diffuse Horizontal Irradiance
diff: Diffuse
dir: Direct
dl: Differential element along the height of a panel
DNI: Direct Normal Irradiance
F: Front
Farm: PV Farm
GHI: Global Horizontal Irradiance
Gnd: Ground
in: Input
M, cell: cell/module
Panel: PV panel
PV: Photovoltaic
STC: Standard testing condition
Total: Total
Z: Zenith